05 Owasp

OWASP Top 10 Assessment — ESPHOME

Company: Anticloud FZ LLE **Standard:** OWASP Top 10 2021

Category Mitigation Status
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A01 Broken Access Control Zero-trust mTLS: every call authenticated PASS
A02 Cryptographic Failures AES-256-GCM at rest, TLS 1.3 in transit PASS
A03 Injection Parameterized queries, no shell exec from user input PASS
A04 Insecure Design Threat model documented, offline-first by design PASS
A05 Security Misconfiguration Single binary: no misconfigurable cloud console PASS
A06 Vulnerable Components SBOM embedded, deterministic builds, pinned deps PASS
A07 Auth & Session Failures mTLS short-lived certs, no session tokens PASS
A08 Software & Data Integrity AIOSS + Provenance chain on every mutation PASS
A09 Security Logging Local observability: all events logged locally PASS
A10 SSRF No outbound HTTP in offline mode, allowlist only PASS

Additional: LLM Top 10 (OWASP LLM01-LLM10 2025)

Category Mitigation Status
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LLM01 Prompt Injection Input sanitization + output validation PASS
LLM02 Insecure Output Handling Structured output validator, schema enforcement PASS
LLM03 Training Data Poisoning Model weights verified via hash, not retrained online PASS
LLM04 Model Denial of Service Token budget enforced, request queuing PASS
LLM05 Supply Chain Vulnerabilities SBOM, deterministic build, pinned GGUF hash PASS
LLM06 Sensitive Info Disclosure Sovereign Memory encrypted, no plaintext logs of prompts PASS
LLM07 Insecure Plugin Design No plugins; all tools compiled into single binary PASS
LLM08 Excessive Agency PAX operates in read-only mode by default PASS
LLM09 Overreliance Human-in-the-loop required for all write operations PASS
LLM10 Model Theft Model runs locally; GGUF never transmitted externally PASS